

Interview preparation sheet

Data Scientist (M/F) · EDF — DPNT / Digital Division / FARNum — Data Ops Team · Ref. 2026-174432 · Saint-Denis (93)

1. The offer at a glance

Title / Reference	Data Scientist (M/F) — Ref. 2026-174432
Company	EDF (SA) — Nuclear and Thermal Plant Directorate (DPNT), Digital Division (DivNum), FARNum entity, Data Ops team
Location	Saint-Denis, Seine-Saint-Denis (93), Île-de-France — France
Contract Type	Permanent Contract — Executive College , M3E grade C range (13-14-15)
Preferred Start Date	01/09/2026
Salary	€46,000 - €57,000 gross annual excluding bonuses, depending on degree and experience
Remote Work	2 days per week
Expected Education	Bachelor's +4/+5, specialisation in Computer Science / Information Systems
Experience	5 years minimum (candidate criteria) — 6 years announced in the desired profile
Target Transversal Competencies	Autonomy, adaptability, leadership, customer focus, results orientation
Professional Family	Information Systems, Telecommunications and Digital — Data Management and Valorisation 2. The Company: EDF and the DPNT Context
Language / Diversity	Offer in French; position open to people with disabilities

Key vocabulary for the company: DPNT = Nuclear and Thermal Park Directorate (operation of power plants) · DivNum = DPNT's Digital Division · FARNum = Rapid Digital Action Force · M3E = grading grid for executives in the Electricity and Gas Industries (IEG) sector. Grade range C (13-14-15) corresponds to a confirmed/expert executive position.

2. The company: EDF and the DPNT context

EDF in Brief

- French integrated electric utility (production, transmission via subsidiaries, distribution, sales), **fully owned by the French State** since its renationalisation in 2023.
- Operates the French nuclear fleet: **57 reactors spread across 18 power plants**, including the EPR of Flamanville, which entered commercial service in 2026.
- 2025 Nuclear Production: **≈ 373 TWh, accounting for ~68% of France's electricity production** — the highest level since 2019.

- Average age of over 40 years, leading to two key programs: the **Grand Carénage** (~€55 billion to extend operation beyond 40 years) and the **EPR2** program (6 to 14 reactors, with the first pair targeted for 2037).

The DPNT and its Digital Division (DivNum)

The DPNT is the department responsible for **operating and maintaining the nuclear and thermal fleet**: safety, availability of units, maintenance, and major industrial construction projects. DivNum aims to **place IT at the heart of industrial performance** within this department: IT system overhaul, cybersecurity, and data management.

Its **4 strategic pillars** (to be cited as-is, they structure the entire entity's discourse):

Standardisation

Cloud^{isation}

Data^{isation}

Industrialisation

The FARNum — Rapid Digital Action Force

- Created to **quickly address the challenges of major industrial projects** within DPNT: an agile, multidisciplinary team, a « catalyst for innovation ».
- Claimed DNA: **pragmatism, results-driven focus, speed of execution, immediate value**, and adaptability to field needs.
- Organised into **3 poles** : *Business & Projects · FAST AI IT · Data Ops*.

The Data Ops team — the one I'm joining

It drives the **operational implementation of Data and AI use cases**, from technical business needs framing to production deployment, with an « industrialised, secure, and value-driven » approach. The role explicitly aims to « *materialise the implementation and use of high-performance, industrial AI* » to meet business needs and DivNum's challenges.

Useful insight: it's neither a R&D team nor a dashboarding team. It's a **delivery** team — end-to-end, with production as the success criterion. This is exactly the angle to adopt throughout the interview.
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3. The role: the 6 main missions

Mission (offer formulation)	What this means concretely
Data approach for use cases: diagnosis, collection, adjustment, valorisation, integration	Retrieve industrial data where it is, map it, and ensure its reliability before any modelling. High data quality.
Identify AI applications during instruction / framing and assess technical feasibility and compliance with Group standards	Internal pre-sales / consulting role: saying no or reformulating a need before committing. Concept of compliance with Group standards = EDF framework, security, sovereignty.
Design end-to-end AI architectures with architects and technical references	Frameworks, infrastructure, model selection, database, patterns, exposure (API), monitoring, alerting, risk management, and drift (data drift) handling.
Deploy, optimize, and pilot in production	Performance, robustness, maintainability, and cost (AI FinOps is explicitly mentioned).
AI technological watch and knowledge sharing	Within FARNum and DivNum — cross-cutting / animation dimension.
Spread best practices: DataOps, MLOps, code review, technical capitalization, use of generative AI by data engineers	Referent posture, not just a contributor. This justifies the C-level framework and 6 years of experience.

4. Expected technical stack — and where I stand

Expected in the offer	Level	My proof to cite
Design, development, and deployment of Data projects in production (6 years)	Solid	Production appetite scoring (BigQuery / Airflow / DataRobot), migration of monthly/weekly pipelines; segmentation deployed on ~4M contracts, including industrialization.
Data Analysis: diagnosis, source mapping, collection, quality improvement	Solid	Data Quality Control rules, resolving BigQuery schema drift (DATEDATETIME, NUMERIC mismatches), SQLAlchemy models, pytest tests on queries.
AI, ML, and generative AI models (LLM, RAG, agents, prompts)	Solid	LLM-based verbatim classification pipeline (internal Mistral gateway, e.g., Qwen3-8B served via vLLM), single/multi-theme evaluation strategy, prompt failure mode diagnosis, extension to a new classification plan of 29 categories.
Python, SQL, and NoSQL, GitHub ecosystem, containerization, API development	Good	Daily Python/SQL; Git (migration of repo to on-prem Azure DevOps); containerization and Cloud Run on GCP. To update: NoSQL and API design (FastAPI).

Best practices for dev (code quality, testing, documentation) and CI/CD	Solid	Multi-layer pytest unit tests, code review, CI/CD with Jenkins, reusable shared library <i>gcs_lib</i>
Cloud architectures (GCP, AWS) and on-premise , Data/AI deployment patterns	Good / to nuance	GCP in production (BigQuery, Vertex AI, Cloud Run, Composer). Real on-premise experience: corporate proxy constraints and SSL inspection, GCS ↔ network share script, on-premise Azure DevOps, internally hosted LLM. <i>AWS = a point to honestly acknowledge.</i>
Agile Methodology	Good	Squad-based operation, iterations, coordination with the Data Engineer on deliverables (Airflow YAML, retraining schedule).
Autonomy, mentorship, team spirit, proactivity, sharing	Solid	Results presentation to Group Marketing Management, restructured deck shifting from a technical angle to a business angle; training and support for business stakeholders.

The point to prepare: the experience. The offer states « 6 years minimum » in the profile and « 5 years » in candidate criteria. Angle to maintain, without justification: count the entire Matmut career (internship, apprenticeship, full-time contract since September 2023, ~4.5 years in the same production environment) and shift the conversation to **density**: multiple models actually in production, end-to-end industrialization, LLM in production — something many 7-year profiles haven't achieved.

5. Remuneration, benefits and conditions

Package

- **Fixed: €46,000 to €57,000 gross annually, excluding bonuses**, depending on degree and professional experience (currency: EUR, annual frequency).
- **Individual performance bonus.**
- **Profit-sharing and employee savings.**
- Remuneration described as «attractive according to profile» — the range is therefore negotiable within its bounds, with the lever being the degree (Bachelor+5 / M2 AI) and experience.

Benefits (subject to eligibility conditions)

- **Preferred rate on electricity and gas** (historical advantage of EDF, to be valued in the calculation of the real package).
- **Family and parental support aids.**
- **Sustainable mobility devices.**
- **Social and cultural activities** (EDF's CCAS).

What the offer highlights as reasons to join

- A varied and interesting job;
- **Remote work 2 days per week** ;
- A **team of human scale** ;
- Specific training for job start and throughout the career;
- Opportunities for advancement within the EDF Group.

Remote work — to clarify in interview: the offer states 2 days per week, but does not specify the schedule (fixed or flexible days), the waiting period after starting, or the possibility of remote work from another region. Three concrete questions to ask: (1) Is the schedule fixed or chosen by the team? (2) Is there a 100% on-site integration period? (3) Is Saint-Denis the only possible point of attachment?

6. Recruitment process

No recruitment steps are mentioned in the offer. The following is the standard EDF process — to be confirmed by the recruiter from the first exchange.

1. **Online application** on the EDF Recrute website (with the offer reference). EDF announces a **first response within 2 weeks**.
2. **Telephone/video exchange with an HR recruiter:** career path, motivation, mobility, salary expectations, availability.
3. **Interview(s) with the operational manager** (presumably the head of the Data Ops department) and sometimes a technical referent/architect: this is where the end-to-end architecture, MLOps, and production dimensions come into play.
4. For managerial positions: **technical test or situational assessment**, and sometimes personality/reasoning questionnaires.

5. **Proposal, then integration** (medical visit, administrative procedures, training for the job start).

Note: EDF hires on nuclear sites may involve **security checks / authorisations** — check if this applies here, as the position is on a tertiary site in Saint-Denis.

7. « Pourquoi postulez-vous chez nous ? » — ma réponse

Short version (30-40 seconds, for an initial HR exchange)

"I've been a Data Scientist for nearly five years at an insurance company, and what drives me isn't the model itself: it's the moment it goes into production and changes a business decision. I've deployed a scoring model in production, a segmentation across the entire portfolio — around four million contracts — and more recently a classification pipeline using LLM. What I'm looking for now is an environment where industrialization is at the core of the job, not a side activity. That's exactly what the Data Ops team at EDF Renewables describes: from defining the need to deploying in production, on high-impact industrial topics. And the 'fast-action force' approach, results-driven, aligns perfectly with how I work."

Long version (2 minutes, for the manager interview) — structured in 4 parts

1. Where I come from and what I've learned. Five years in the same company, on the same production environment, from internship to permanent contract. I've done the full journey: building a model, then learning what 'keeping it alive' really means — shifting a score from a monthly to a weekly rhythm, managing schema drifts, writing tests, data quality checks, and troubleshooting when a pipeline fails. That's what pushed me toward an engineering profile rather than pure analysis.

2. Why I'm leaving. Without criticizing: I've reached the limits of what I can learn about industrialization where I am. I want to join a team where this is their core business, with architects and technical references facing me, and a framework that demands excellence in production deployment.

3. Why EDF, and why this team specifically. Three concrete reasons:

- *The topic.* The industrial performance of the nuclear and thermal fleet is high-stakes data, with reliability and security constraints not found everywhere. Moving from a marketing use of data to an industrial one is the scale shift I'm interested in.
- *The team's positioning.* Data Ops handles use cases from definition to production, with an industrialized and secure approach. It's literally the role I'm trying to fill today, but in a framework where it's the official mission.
- *The transformation context.* The four pillars of DivNum — standardization, cloudification, datafication, and industrialization — describe a trajectory I can contribute to immediately, particularly in standardizing MLOps practices and sharing best practices, which is explicitly part of the missions.

4. What I bring from day one. Real production experience in a constrained context: cloud GCP, but also on-premise with corporate proxies and SSL inspection; an internally served LLM rather than a public API — which directly addresses sovereignty and Group compliance standards. And a practice of communication: I've presented my results to a business director, rebuilding the narrative from a technical angle to a business one. Pedagogy and sharing are part of your expectations, and it's a part of the job I enjoy.

What NOT to say: relocation to Île-de-France must not appear as *the* reason for the application; nor should compensation. If mobility is asked about: simple and honest answer — « it's a mobility project I've already anticipated; Île-de-France is one of my target areas. » And if asked why leaving the current job: focus on the professional project, never on salary frustration.

8. Likely questions and my angles

Expected question	My angle
« Tell me about your biggest project »	The end-to-end production score (not segmentation): end-to-end, production constraints, pace migration, feedback loop and conditional retraining, data quality control (DQC), testing. This is the project that meets the expectations of this role. Segmentation comes later, for scale (~4 million contracts). end-to-end, production constraints, pace migration, feedback loop and conditional retraining, data quality control (DQC), testing. This is the project that meets the expectations of this role. Segmentation comes later, for scale (~4 million contracts). « How do you deploy an AI model end-to-end? » Unfold the chain as the offer names it: infrastructure framework, model choice, database, exposure pattern (batch/API), monitoring, alerting, drift management, cost. Use their vocabulary.
« What have you done in generative AI? »	The verbatim pipeline: choice of an internally served LLM, evaluation mono vs multi-theme, prompt failure modes (neutrality bias, false positives), transition to a classification plan of 29 categories. Emphasize evaluation, that's what distinguishes industrial use from a demo.
« How do you handle model drift in production? »	Data drift vs concept drift, monitoring input distributions and performance, alert thresholds, conditional retraining (already implemented), rollback and versioning. « Nuclear energy, that's not your sector » Own it: I don't bring nuclear fleet expertise, but I bring a method for deployment in a regulated and constrained environment. Insurance shares the requirements for traceability, auditability, and compliance. ranking of 29 categories. Emphasise the assessment, that is Position yourself at the top end of the range (Master's degree in AI + production experience), recalling that the package includes performance bonus and profit-sharing. Do not mention a figure below €46K.
« AWS? NoSQL? »	Don't overplay it: GCP in production, transferable principles, and existing curiosity through the self-training program (containerization, Kubernetes, Infrastructure as Code, observability) and the Google ML Engineer pathway.
9. My questions to ask	1. How is the Data Ops team currently composed — how many Data Scientists, Data Engineers, and is there a dedicated MLOps team? 2. What are the 2 or 3 top use cases for the team over the next 6 months? 3. What is the technical foundation in place: cloud, on-premise, or hybrid? What platform is used for model training and serving? traceability, auditability, and compliance.
"Your claims?"	Positioning yourself at the top end (Bac+5 M2 AI + production experience), while noting that the package includes performance bonus and profit-sharing. Do not provide a figure below 46 K€.
"AWS? NoSQL?"	Do not overstate: GCP in production, transferable principles, and curiosity already fostered through the self-training program (containerization, Kubernetes, IaC, observability) and the Google ML Engineer pathway.

9. My questions to ask

1. How is the current Data Ops team composed — how many Data Scientists, Data Engineers, is there a dedicated MLOps?
2. What are the 2 or 3 top use cases for the team over the next 6 months?
3. What is the technical foundation in place: cloud, on-premise, or hybrid? Which platform for training and serving the models?
4. How are the « Group Standards » for AI compliance concretely implemented — is there an evaluation and validation framework for the models?
5. Which LLMs are used and how are they deployed (sovereignty, on-premise hosting)?
6. How is time allocated between upstream framing, development, and execution of products already in production?

7. What is the relationship between the Data Ops team and the FAST AI IT team?
8. Work-from-home schedule and modalities (fixed or flexible days, waiting period before starting the job)?
9. What are the next steps in the process and under what timeline?

10. Pre-interview checklist

- ☐ Review the offer and be able to cite the **4 pillars** of DivNum and the **3 teams** of FARNum.
- ☐ Prepare 3 STAR-format stories: production score · LLM classification · a resolved production incident.
- ☐ Quantify each project (volumes, pace, gains, deadlines) — the team positions itself as 'results-driven'.
- ☐ Prepare a 60-second response on 'data quality': it is the first listed mission.
- ☐ Verify and state your salary expectations confidently.
- ☐ Identify the route to the Saint-Denis site and plan extra time.
- ☐ Print or have on hand the list of questions to ask.

Sources: EDF job offer ref. 2026-174432 (PDF provided) · edf.fr — recruitment steps · conaissancedesenergies.org — French nuclear fleet 2026. The recruitment process timeline is not detailed in the offer and remains to be confirmed with the recruiter.